

Football Player Analytics: A CRISP-DM Approach to Scouting and Performance Analysis

Team Members

- Nada Gamal
 - Abdullah Mohamed
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1. Introduction

The football industry has increasingly adopted data analytics to improve scouting, recruitment, and tactical decision-making. Modern clubs collect large volumes of player statistics, but extracting meaningful insights from high-dimensional football data remains a major challenge.

This project applies the CRISP-DM methodology to analyze football player statistics from the 2022–2023 season across the top European leagues. The objective is to transform raw statistical data into actionable scouting insights using multiple data mining techniques.

The project focuses on:

- Identifying hidden player playstyles
 - Predicting offensive performance
 - Classifying player scoring ability
 - Visualizing player similarity in reduced-dimensional space
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2. Business Understanding

Business Problem

Football clubs often operate under financial constraints and must identify undervalued talent efficiently. Traditional scouting methods may overlook players whose underlying statistics indicate strong potential.

The challenge is to use data mining techniques to:

- Discover meaningful player profiles
- Predict offensive output
- Support recruitment and tactical analysis

Business Objectives

The main objectives of this project are:

1. Classify players as High Scorers or Low Scorers based on performance statistics.

2. Cluster players into statistical playstyle groups independent of traditional positions.
 3. Predict goal-scoring output using attacking metrics.
 4. Build an interactive dashboard that allows decision-makers to explore player insights visually.
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3. Dataset Understanding

Dataset Description

The dataset used in this project is:

2022–2023 Football Player Stats Dataset

The dataset contains player statistics collected from the top five European football leagues.

Dataset Size

- Total Players: 2,689
- Features: More than 120 statistical attributes

Data Types

Identifying Features

- Player Name
- Nation
- Squad
- Competition
- Position

Numerical Features

- Goals
- Assists
- Minutes Played
- Shot-Creating Actions (SCA)
- Progressive Passes
- Tackles Won
- Passing Statistics
- Defensive Metrics

4. Data Preparation

Several preprocessing steps were applied to improve data quality and prepare the dataset for machine learning models.

Data Cleaning Steps

Filtering Low-Minute Players

Players with fewer than 300 minutes played were removed to reduce statistical noise and unreliable performance patterns.

Handling Missing Values

Missing numeric values were filled with zero values.

Feature Scaling

StandardScaler was applied to normalize numerical features. This step was especially important because distance-based algorithms such as K-Means and PCA are sensitive to feature scale differences.

Feature Engineering

A new target variable called Scorer_Class was created to classify players into:

- High Scorer
- Low Scorer

This classification was based on the median number of goals scored.

5. CRISP-DM Methodology

CRISP-DM Phase	Activities Performed
Business Understanding	Defined scouting and performance analysis objectives
Data Understanding	Explored correlations, distributions, and player statistics
Data Preparation	Cleaned data, scaled features, engineered target variables
Modelling	Applied Classification, Clustering, Regression, and PCA
Evaluation	Evaluated models using accuracy, regression metrics, and clustering analysis

CRISP-DM Phase	Activities Performed
Deployment	Built an interactive Streamlit dashboard for decision-makers

6. Modelling Techniques

6.1 Classification — Random Forest

A Random Forest classifier was used to classify players into High Scorer and Low Scorer categories.

Purpose

To identify statistical patterns associated with offensive productivity.

Result

The model successfully learned the characteristics commonly associated with goalscoring players.

Misclassified players may indicate:

- Players with strong underlying attacking metrics but poor finishing
 - Overperforming players scoring above expectations
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6.2 Clustering — K-Means

K-Means clustering was used to group players into statistical archetypes.

Purpose

To discover hidden player profiles without predefined labels.

Result

The algorithm grouped players into four distinct clusters representing different football playstyles.

The clustering results demonstrated that modern football roles are increasingly hybrid and not limited to traditional positions.

6.3 Regression — Linear Regression

Linear Regression was applied to predict goal output using attacking metrics such as:

- Shot-Creating Actions (SCA)
- Shots

- Shots on Target

Purpose

To estimate offensive output using measurable attacking actions.

Result

A strong linear relationship was observed between Shot-Creating Actions and Goals scored.

Players significantly above the regression trendline appeared more clinically efficient than expected.

6.4 Dimensionality Reduction — PCA

Principal Component Analysis (PCA) was applied to reduce the dimensionality of the dataset.

Purpose

To simplify over 100 statistical features into a 2D representation for visualization and similarity analysis.

Result

PCA successfully revealed visual player groupings and performance gradients between defensive and attacking player profiles.

7. Evaluation

The models were evaluated using multiple performance metrics.

Classification Evaluation

- Accuracy Score
- Confusion Matrix
- Classification Report

Regression Evaluation

- R^2 Score
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

Clustering Evaluation

- Elbow Method
- Silhouette Score

The evaluation results demonstrated that the selected models produced meaningful and interpretable football insights.

8. Dashboard Deployment

An interactive dashboard was developed using Streamlit as the deployment layer of the project.

Dashboard Features

- Global league filtering
- Interactive PCA visualization
- Player clustering analysis
- Classification insights
- Regression trend analysis
- Business insights panel

The dashboard enables users to explore football analytics interactively and supports scouting-oriented decision-making.

9. Key Findings

1. Modern football player roles are increasingly hybrid rather than strictly position-based.
 2. Shot-Creating Actions were among the strongest indicators of offensive output.
 3. Clustering techniques successfully identified hidden player archetypes.
 4. PCA enabled intuitive visual exploration of player similarity.
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10. Limitations

Although the project produced meaningful insights, several limitations remain:

- Tactical systems and team strategies are not represented in the dataset.
 - The dataset only covers one season.
 - External factors such as injuries and teammate quality were not included.
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11. Future Work

Future improvements may include:

- Integrating Expected Goals (xG) data
 - Multi-season longitudinal analysis
 - Advanced event-based tracking data
 - Deep learning models for player similarity analysis
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12. Conclusion

This project successfully demonstrated the application of the CRISP-DM framework in sports analytics using football player statistics.

By combining classification, clustering, regression, and dimensionality reduction techniques, the project transformed raw football statistics into interpretable scouting insights.

The final Streamlit dashboard serves as an interactive decision-support tool for player analysis, recruitment, and tactical evaluation.